THE THEORY OF MEASURES AND INTEGRATION

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Abstract. Lebesgue's integral patches up many of the defects of Riemann's integral, at the expense of the need to develop a notion of *measure*. These notes offer a broad and accelerated overview of the Lebesgue integral and measure theory with several exercises, open ended questions, and examples.

1. Lecture 1: Some Motivation, Basic Definitions and Results

1.1. Motivation. Length, Area, and Volume

We start with a simple question of *what is length?* We should be able to get some mathematical model for it. The standard objects of length measurements are one-dimensional, so it is fair to restrict ourselves to working in $\mathbb{R}$. In this setting, the objects of interest are, at their simplest, line segments, which are in turn naturally modeled by *intervals*.

So if we have an interval $[a, b] \subseteq \mathbb{R}$, it is then quite immediate to suggest its length:

$$\text{length of } [a, b] = b - a$$

1.2. **Basic Definitions and Results.** Our first definition formalizes the collections of sets that we can measure.

Definition 1.1. Given a set Ω, a *σ-algebra of subsets of* Ω is a family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that

(i) $\emptyset, \Omega \in \mathcal{F}$,
(ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and
(iii) if $\{A_k\}_{k\in\mathbb{N}} \subseteq \mathcal{F}$ then $\bigcup_{k\in\mathbb{N}} A_k \in \mathcal{F}$.

We'll say a pair $(\Omega, \mathcal{F})$ is a *measurable space*.

So σ-algebras are closed under countable unions, but they are also closed under countable intersections!

Question 1. *Show that if* $\{A_k\}_{k\in\mathbb{N}} \subseteq \mathcal{F}$ *then*

$$\bigcap_{k\in\mathbb{N}} A_k \in \mathcal{F}.$$

Let's pause here for a moment to ask, what kinds of σ-algebras are there?

Example 1.2. If Ω is a set, then $\mathcal{P}(\Omega)$ is a σ-algebra.

Actually *all* finite σ-algebras are built from these. (See Question 10)

But it would be dull if all σ-algebras were just power sets.

Definition 1.3. Given a set Ω and a family of subsets $\mathcal{S} \subseteq \mathcal{P}(\Omega)$, we define the *$\sigma$-algebra generated by* $\mathcal{S}$ to be the smallest σ-algebra containing $\mathcal{S}$. That is,

$$\sigma(S) := \bigcap_{\substack{\mathcal{F}\subseteq\mathcal{P}(\Omega) \\ \mathcal{F} \text{ is a } \sigma\text{-algebra}}} \mathcal{F}.$$

It is a simple exercise to show that these generated σ-algebras are indeed σ-algebras. But how are these useful? Well one example of when we deal with a set of subsets that is not inherently a σ-algebra, is the setting of a topological space.

Example 1.4. Let (X, τ) be a topological space. Then the σ-algebra generated by the topology (i.e. all the open sets), *Borel σ-algebra*, which we will denote by

$$\mathcal{B}(X, \tau) := \sigma(\tau).$$

For example, take $\mathbb{R}$ with its usual (metric) topology. Then the Borel σ-algebra is a non-trivial (i.e. not $\mathcal{P}(\mathbb{R})$) σ-algebra containing all open subsets of $\mathbb{R}$. We will revisit this non-trivialness after introducing the Lebesgue measure to show, using the Axiom of Choice to construct "Vitali sets", which are non-Borel subsets of $\mathbb{R}$.

Question 2. *Show that $\mathcal{B}(\mathbb{R})$ contains all closed sets, all F_σ sets, and all G_δ sets.*

Curiosity 1. *What is the cardinality of $\mathcal{B}(\mathbb{R})$? Seems like it contains a lot of sets, so it could be that $|\mathcal{B}(\mathbb{R})| = |\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}}$.*

However, it definitely does not contain all sets. So perhaps we have $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$?

If the latter is the case, that would be quite shocking, as all imaginable or constructable sets appear to be Borel, yet there is a whole size of infinity larger than the Borel sets composed of the unimaginable ones!

Question 3. *Show that the Borel subsets of $\mathbb{R}$ (with its metric topology) are also generated by the following collections*

(i) $\{(a, b] : -\infty \le a < b < \infty\}$,
(ii) $\{(a, \infty) : a \in \mathbb{R}\}$
(iii) $\{(a, b) : a < b \in \mathbb{R}\}$

In general, we have the following. The proof is a challenging exercise, but not necessary to get stuck on.

Fact 1. *If $\mathcal{F}$ is an infinite σ algebra, then $\mathcal{F}$ is uncountable.*

Definition 1.5. Given a measurable space $(\Omega, \mathcal{F})$, a (positive) *measure* on $\mathcal{F}$ (or on Ω if $\mathcal{F}$ is understood) is a function $\mu : \mathcal{F} \to [0, \infty]$ such that

(i) $\mu(\emptyset) = 0$, and
(ii) if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

A triple $(\Omega, \mathcal{F}, \mu)$ where $\mathcal{F}$ is a σ-algebra of subsets of Ω and μ is a measure on $\mathcal{F}$ will be called a *measure space.*

Our notions of length, area, and probability are all examples of measures, but their formal constructions will have to wait until we have some more tools. For now, let's see a couple of simple but elegant examples of measures that will end up bridging the gap between sums and integrals.

Example 1.6. Take $\Omega = \mathbb{N}$ and $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define the *counting measure on* $\mathbb{N}$, $\mu_{\text{count}} : \mathcal{P}(\mathbb{N}) \to [0, \infty]$ by

$$\mu_{\text{count}}(S) = |S|.$$

It is not hard to see that this satisfies all requirements of our definition.

In fact, we can define this same counting measure on any set, which we may encounter later. But the counting measure on $\mathbb{N}$ will be of particular importance to us.

One may reasonably call this a zero-dimensional volume measure, like length is to one-dimension and area is to two-dimensions!

Example 1.7. More fundamentally, for any Ω and any $a \in \Omega$, we have the *Dirac mass at a* measure, which is defined by

$$\delta_a(S) = \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{otherwise} \end{cases}.$$

If measures are showing us how volume or mass is distributed, then this one says that all of it is concentrated at the point a. It is a point-mass!

Dirac masses are the simplest measures we can have but they are surprisingly useful.

Question 4. *The two measures we saw above are connected in a very direct way. Show that*

$$\mu_{count} = \sum_{n=1}^{\infty} \delta_n.$$

Question 5. *Motivated from the previous question, show that adding two measures results in a measure.*

Not all linear combinations of measures are permissible. Which ones are?

Such a space is often called a convex cone.

Question 6. *Show that every measurable space can be upgraded to a measure space with some measure.*

Curiosity 2. *Can we classify the set of measures somehow? Maybe in the simplest setting, say $|\Omega| < \infty$. Can we characterize all measures on $(\Omega, \mathcal{P}(\Omega))$? In this setting, if F is a sub-σ-algebra of $\mathcal{P}(\Omega)$ and μ is a measure on $(\Omega, \mathcal{F})$, can μ be extended to $\mathcal{P}(\Omega)$ for free?*

We also have measures associated to length and area, as we started out. But showing their existence is a bit more involved. For now, we delay this problem to a future section.

The requirement for the additivity of measure that we have disjoint sets is not a big limitation, for we can always *disjointify* a family, and at least get sub-additivity.

Question 7. *For a given family of sets $\{A_k\}_{k \in K}$, show that there is a disjoint family $\{B_k\}_{k \in K}$ such that*

$$\bigcup_{k=1}^{n} A_k = \bigcup_{k=1}^{n} B_k.$$

Question 8. *Show that, with the setup above,*

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(B_k).$$

Argue that for any countable collection $\{A_k\}_{k \in K} \subseteq \mathcal{F}$, we have sub-additivity of measure:

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} \mu(A_k).$$

We also quickly get using the same ideas the following computational tool for dealing with non-disjoint sets:

Theorem 1.8 (Inclusion-Exclusion)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $A, B \in \mathcal{F}$. Then*

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Remark 1.9. One other objection one might have to the definition of measure is that we can seemingly define it equally fine for just finite unions. For example, one can define a *finitely-additive measure* on an *algebra* $\mathcal{F}$ of subsets of Ω as a map $\mu : \mathcal{F} \to [0, \infty]$ satisfying

(i) $\mu(\emptyset) = 0$, and
(ii) If $A_1, \ldots, A_n \in \mathcal{F}$ are pairwise disjoint, then

$$\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n).$$

It is immediate that all measures in our original sense are finitely additive in this sense (why?). So, what benefit do we get from allowing for infinite unions?

A short possible justification is that by allowing countably infinities, we can deal with notions of limits and continuity, which vastly expands our scope. These kinds of arguments will be crucial for developing Lebesgue's integral later on.

The following is a result that is unique to countably-additive measures.

Theorem 1.10 (Continuity of Measure)**.** *If $\{A_k\}_{k\in K} \subseteq \mathcal{F}$ with $A_k \subseteq A_{k+1}$ for all $k \in \mathbb{N}$ then*

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

Furthermore, if $\{A_k\}_{k\in K} \subseteq \mathcal{F}$ with $A_k \supseteq A_{k+1}$ for all $k \in \mathbb{N}$, and if $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

Question 9. *Show that without the assumption that $\mu(A_1) < \infty$ in the second part of the theorem above, the statement may not be true.*

1.3. Exercises.

Question 10. *We will show that all finite measurable spaces are of the form $(\Omega, \mathcal{P}(\Omega'))$. Let $(\Omega, \mathcal{F})$ be a measurable space where $|\mathcal{F}| < \infty$.*

- i *Say a nonempty subset $A \in \mathcal{F}$ is minimal if $B \in \mathcal{F}$ and $B \subseteq A$ implies that $B = \emptyset$ or $B = A$. Show that the collection of minimal elements of $\mathcal{F}$ partitions Ω (i.e. they are all disjoint and union to Ω.*
- ii *Find an appropriate set Ω' and a function $f : \Omega \to \Omega'$ such that $\hat{f} : \mathcal{F} \to \mathcal{P}(\Omega')$, defined by $\hat{f}(A) := f(A)$ is an order-isomorphism of $\mathcal{F}$ to $\mathcal{P}(\Omega')$.*

Question 11. *Prove Fact 1. (Hint: $|\mathcal{P}(\mathbb{N})| > |\mathbb{N}|$)*

Question 12. *Show that the σ-algebra generated by singletons on $\mathbb{R}$ is not $\mathcal{P}(\mathbb{R})$.*

Question 13. *Show that there are finitely-additive measures that are not countably-additive. (Hint: such a measure must fail the continuity theorem for some collection of sets)*

2. Lecture 2: Lebesgue's Integral

Motivation: balancing act

Question 14. *If we want to have our cake and eat it too, what functions satisfy having every preimage of an interval being also an interval?*

2.1. Permissible Functions.

To allow for the chaos in Lebesgue's idea for the integral, we need to demand a certain property of our functions. In particular, for f to be integrable, a necessary condition is that the pre-images of intervals must have the ability to be measured. This condition is called *measurability.*

Definition 2.1. Given a measure space $(\Omega, \mathcal{F}, \mu)$, a function $f : \Omega \to \mathbb{R}$ is *measurable* if for all $B \in \mathcal{B}(\mathbb{R})$,

$$f^{-1}(B) \in \mathcal{F}.$$

For convenience, we denote the set of measurable functions on Ω by $\mathrm{Meas}(\Omega, \mathbb{R})$. We will also sometimes restrict ourselves to positive measurable functions (that is, $f : \Omega \to [0, \infty)$ measurable), and we will denote this set by $\mathrm{Meas}^+(\Omega, \mathbb{R})$.

Question 15. *Show that all continuous functions $f : \mathbb{R} \to \mathbb{R}$ are Borel measurable. (Hint: It suffices to look at pre-images of open sets. Why?)*

Theorem 2.2. *If* $(\Omega, \mathcal{F})$ *is a measurable space,* f, g *measurable and* $\lambda \in \mathbb{R}$ *then we have*

(i) λf *is measurable*
(ii) $f + g$ *is measurable*
(iii) fg *is measurable*
(iv $|f|$ *is measurable*
(v) $\max\{f, g\}$ *and* $\min\{f, g\}$ *are measurable.*

Proof. We will show (ii), a surprisingly slippery argument ((iii) is similar), and leave the rest as an exercise for the reader which are a bit more direct.

□

Remark 2.3. Properties (i), (ii), and (iii) in the theorem above say that $\mathrm{Meas}(\Omega, \mathbb{R})$ is an algebra. Property (v) guarantees us that it is also a lattice. So $\mathrm{Meas}(\Omega, \mathbb{R})$ has a decently rich algebraic structure to it.

In addition, the question preceding the theorem above shows us that when $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then

$$C(\mathbb{R}, \mathbb{R}) \subseteq \mathrm{Meas}(\mathbb{R}, \mathbb{R}).$$

This containment is strict. There is a much richer collection of measurable functions than continuous ones.

Example 2.4. A simple example of discontinuous measurable functions are the *characteristic functions.* Let $(\Omega, \mathcal{F})$ be a measurable space, and let $E \in \mathcal{F}$. Then define the associated characteristic function, χ_E as follows.

$$\chi_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E \\ 0 & \text{otherwise} \end{cases}.$$

It is easy to see that such a function is measurable. Indeed, let B be any Borel subset of $\mathbb{R}$, then we have

$$\chi_E^{-1}(B) = \begin{cases} \Omega & \text{if } 0, 1 \in B \\ E & \text{if } 1 \in B \text{ and } 0 \notin B \\ E^c & \text{if } 1 \notin B \text{ and } 0 \in B \\ \emptyset & \text{otherwise} \end{cases}.$$

So for example, if $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and $E = \mathbb{Q}$, then $\chi_{\mathbb{Q}}$, the characteristic function of the rational numbers, also known as the Dirichlet function, is everywhere discontinuous but measurable.

Curiosity 3. *There are many questions we can ask about the nature of measurable functions to understand them better. For starters, what is* $|Meas(\Omega, \mathbb{R})|$*?*

Also, how different are measurable functions from continuous functions, really? Though the Dirichlet function is discontinuous everywhere, it seems that if we changed its function values on a set of zero measure (the rationals), we could get a continuous function.

Does $Meas(\mathbb{R}, \mathbb{R})$ *have any kind of natural metric or topology associated to it, to tell us when measurable functions are close? (We will soon see that in fact this space is closed under point-wise limits!)*

Let's get on with defining our integrals. We proceed in stages, starting with the simplest functions, and using these to build up to the most general cases. For the following development, fix a measure space $(\Omega, \mathcal{F}, \mu)$.

Step One: Characteristic Functions

Let $E \in \mathcal{F}$ and consider its characteristic function χ_E. If we apply Lebesgue's approach to defining the integral, we come up with the following reasonable definition for the integral of E.

$$\int \chi_E \, d\mu := \mu(E).$$

So far so good, but characteristic functions are a very limited class of functions. We can improve this immediately by extending this definition linearly, as follows.

Step Two: Simple Functions

Definition 2.5. A measurable function $\varphi : \Omega \to \mathbb{R}$ is called *simple* if $|\operatorname{Ran} f| < \infty$.

Question 16. *Show that simple functions are exactly the linear combinations of characteristic functions.*

Let φ be simple with $\operatorname{Ran} \varphi = \{r_1, \ldots, r_n\}$. Then define

$$\int \varphi \, d\mu := \sum_{k=1}^{n} r_k \cdot \mu(\varphi^{-1}(\{r_k\})).$$

This is still giving us a seemingly very small class of functions to work with, but now that we have these simple functions, we can use these to approximate more arbitrary functions. We start with the positive ones.

Step Three: Positive Functions

Even with Riemann's approach, we approximated the area under a curve with simple functions which were constant on subintervals. Lebesgue's idea of chopping up the range allows us to use simple functions which are constant on measurable partitions, expanding the class of approximation functions significantly. But the following definition closely parallels the Riemann approach.

Definition 2.6. If $f : \Omega \to [0, \infty)$ is measurable define

$$\int f d\mu = \sup_{\substack{\varphi \text{ simple} \\ \varphi \leq f}} \int \varphi \, d\mu.$$

Question 17. *Could we have defined this as an infimum instead, over all simple functions larger than f?*

Step Four: Generalizing

To extend this integral to a more general class of measurable functions, we use the ability to write a function in terms of positive ones. The following lemma takes care of this.

Lemma 2.7. *If $f \in Meas(\Omega, \mathbb{R})$ then there exist $f_+, f_- \in Meas^+(\Omega, \mathbb{R})$ such that*

$$f = f_+ - f_-$$

Proof. □

Now one might naturally come up with the following extension of the integral to f:

$$\int f d\mu := \int f_+ d\mu - \int f_- d\mu.$$

This is indeed how we do this, however we offer one caveat. Namely, the integrals may not have been finite, so there is potential in the above that we are left with the paradoxical

$$\int f d\mu = \infty - \infty.$$

To avoid this, we simply stipulate that for the integral of f to be well defined, then at most one of $\int f_+ d\mu$ and $\int f_- d\mu$ is infinite.

Definition 2.8. We will say a measurable function f is *integrable* if

$$\int |f| d\mu = \int f_+ d\mu + \int f_- d\mu < \infty.$$

Denote by $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ the space of integrable functions on $(\Omega, \mathcal{F}, \mu)$.

Now that the integral is properly defined, let's take a moment to verify some basic properties. The following lemma says that the integral is both *monotone* and *linear*.

Lemma 2.9. *If f, g are integrable with $f \leq g$, then*

$$\int f d\mu \leq \int g d\mu.$$

Furthermore, if $\lambda \in \mathbb{R}$, we have

$$\int (\lambda f + g) d\mu = \lambda \int f d\mu + \int g d\mu.$$

The Lebesgue integral respects limits very nicely too. We start with measurability.

Theorem 2.10. *If $\{f_n\}_{n \in N}$ is a sequence of measurable functions which converges point-wise to f, then f is measurable.*

Theorem 2.11 (Monotone Convergence Theorem)**.** *If $\{f_n\}_{k \in \mathbb{N}}$ is a family of non-negative, measurable functions such that $f_{n+1} \geq f_n$ and $f(x) := \sup_{n \in \mathbb{N}} f_n(x)$, then*

$$\int f d\mu = \lim_{n \to \infty} \int f_k d\mu.$$

This discussion has been largely abstract. Let's see if we can compute some examples!

Example 2.12. Let Ω be any set and let $\mathcal{F} = \mathcal{P}(\Omega$. Fix $a \in \Omega$ and consider the Dirac mass δ_a. Then integration against δ_a has a surprising form: for any $f \in \mathrm{Meas}(\mathbb{R}, \mathbb{R})$, we have

$$\int f d\delta_a = f(a).$$

So integration of a function against the Dirac mass at a is just evaluation of f at a!

This is not too difficult to show directly from the definition, see if you can.

Example 2.13. Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, and take the counting measure μ_{count}. Then, for $f : \mathbb{N} \to \mathbb{R}$ integrable with respect to μ_{count}, we have

$$\int f d\mu_{\text{count}} = \sum_{n=1}^{\infty} f(n).$$

Furthermore, f is integrable with respect to μ_{count} if and only if its corresponding series is absolutely summable.

So every time you evaluated a function or took a summation, you were doing integrals!

2.2. Exercises.

Question 18. *Show that if $f : [a, b] \to \mathbb{R}$ is continuous, then it is Lebesgue integrable, and furthermore its Lebesgue integral equals its Riemann integral.*

See if you can generalize this to all Riemann integrable functions on $[a, b]$.

Question 19. *Show that $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ is a vector space equipped with a semi-norm*

$$\|f\|_1 = \int |f| d\mu.$$

Why is it not a norm in general?

When taking a quotient of $\mathcal{L}^1$ by the relation $f \sim g$ if $\|f - g\|_1 = 0$, we get the quotient space

$$L^1(\Omega, \mathcal{F}, \mu) = \mathcal{L}^1(\Omega, \mathcal{F}, \mu)/\sim .$$

It is a nice fact that this turns out to be a Banach space.

3. Lecture 3: The Riesz-Markov-Kakutani Representation Theorem

4. Lecture 4: Radon-Nikodym Derivatives